

SIMATS ENGINEERING
CO1 - ASSESSMENT TOOL 1
Analytical Problem Solving

COURSE CODE: MLA0405

COURSE NAME: Deep Learning

Duration: 1 Hour | Marks: 50

QUESTIONS: 5 | Total Marks: 50

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Code of Conduct:

I P.Bhanu Prakash / 192425263 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____ Name of the student: _____

Annexure A: Analytical Problem Solving (Questions 1 to 5)

Question 1: MLE for Coin Toss Outcomes

A coin is tossed 10 times and the outcomes are: H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

1. Problem Formulation & Parameter Identification

- **Observed Data:** Total tosses $n = 10$. Sequence: H, H, T, H, H, H, T, H, H, T.
- **Counts:** Number of Heads (k) = 7, Number of Tails ($n - k$) = 3.
- **Parameter to Estimate:** Probability of getting Heads p in $[0, 1]$.

2. Mathematical Derivation using Maximum Likelihood Estimation

Assuming independent Bernoulli trials, the likelihood function $L(p)$ for obtaining k heads in n tosses is:

$$L(p) = p^k * (1 - p)^{(n - k)} = p^7 * (1 - p)^3$$

Taking the natural logarithm to obtain the log-likelihood $\log L(p)$:

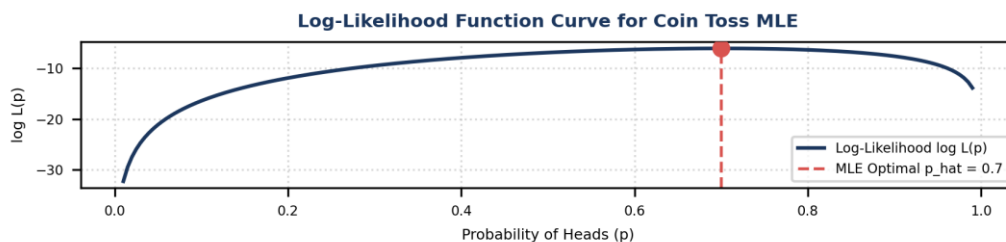
$$\log L(p) = 7 * \log(p) + 3 * \log(1 - p)$$

To find the MLE estimate p_{hat} , take the derivative with respect to p and set it to zero:

$$d(\log L)/dp = (7 / p) - (3 / (1 - p)) = 0$$

$$7 * (1 - p) = 3 * p \Rightarrow 7 - 7p = 3p \Rightarrow 10p = 7 \Rightarrow p_{\text{hat}} = 7 / 10 = 0.7$$

3. Log-Likelihood Optimization Visualization



4. Final Result & Conclusion

Final Solution: The Maximum Likelihood Estimator for the probability of getting Heads is $p_{\text{hat}} = 0.7$ (or 70%).

Question 2: MLE for Student Exam Pass Probability

Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

1. Problem Identification & Bernoulli Modeling

- Sample Size (n):** Total students tested $n = 50$.
- Success Count (k):** Number of students who passed $k = 42$.
- Distribution Model:** Each student performance $X_i \sim \text{Bernoulli}(p)$, where $X_i = 1$ (Pass) with prob p , and $X_i = 0$ (Fail) with prob $1 - p$.

2. Likelihood Formulation & Optimization

The general MLE for a Bernoulli parameter p given n trials and k successes is derived as:

$$L(p) = C(n, k) * p^k * (1 - p)^{(n - k)}$$

$$\log L(p) = \log C(n, k) + k * \log(p) + (n - k) * \log(1 - p)$$

Differentiating with respect to p and equating to zero yields:

$$d(\log L)/dp = (k / p) - ((n - k) / (1 - p)) = 0 \Rightarrow \hat{p} = k / n$$

Substituting given parameters: $\hat{p} = 42 / 50 = 0.84$

3. Bernoulli MLE Estimation Summary

Bernoulli MLE Estimation Workflow

Total Students (n) = 50 | Pass Count (k) = 42
 $\hat{p} = k / n = 42 / 50 = 0.84$ (84% Pass Probability)

4. Final Answer

Final Solution: The MLE estimate for the probability that a student passes the exam is $\hat{p} = 0.84$ (84%).

Question 3: MLE for Defective Light Bulb Rate

A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.

1. Parameter Specification

- **Total Sampled Bulbs (n):** 100
- **Defective Bulbs (k):** 8
- **Non-Defective Bulbs (n - k):** 92

2. Derivation of MLE for Defect Rate (p)

Let p be the probability that a randomly chosen bulb is defective. Under i.i.d. Bernoulli sampling:

$$L(p) = p^8 * (1 - p)^{92}$$

$$\log L(p) = 8 * \log(p) + 92 * \log(1 - p)$$

Setting derivative $d(\log L)/dp = 0$:

$$(8 / p) - (92 / (1 - p)) = 0 \Rightarrow 8 * (1 - p) = 92 * p \Rightarrow 8 = 100 * p \Rightarrow p_{\text{hat}} = 8 / 100 = 0.08$$

3. Defect Rate Calculation Breakdown

Quality Control Defect Rate MLE

Sample Size = 100 | Defects = 8
 $p_{\text{hat}} = \text{Defective Count} / \text{Total Sample} = 8 / 100 = 0.08$ (8%)

4. Final Answer

Final Solution: The MLE for the probability that a bulb is defective is $p_{\text{hat}} = 0.08$ (8%).

Question 4: MLE for Biased Die Rolling Probability

A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.

1. Problem Parameters

- **Total Rolls (N):** 60
- **Occurrences of Number 6 (x):** 18
- **Event Definition:** Success = Rolling a 6; Failure = Rolling any other number (1-5).

2. Mathematical Solution

Let $p = P(X = 6)$. Modeling as binomial/Bernoulli trial likelihood:

$$L(p) = p^{18} * (1 - p)^{42}$$

$$\log L(p) = 18 * \log(p) + 42 * \log(1 - p)$$

Maximizing log-likelihood:

$$d(\log L)/dp = (18 / p) - (42 / (1 - p)) = 0$$

$$18 * (1 - p) = 42 * p \Rightarrow 18 - 18p = 42p \Rightarrow 60p = 18 \Rightarrow p_{\text{hat}} = 18 / 60 = 0.30$$

3. Die Frequency vs Fair Probability Comparison

Biased Die MLE Result

Total Rolls = 60 | Face '6' Frequency = 18
 $p_{\text{hat}} = 18 / 60 = 0.30$ (vs Fair Die Expected $1/6 = 0.167$)

4. Final Answer

Final Solution: The estimated probability of rolling a 6 using MLE is $p_{\text{hat}} = 0.30$ (30%).

Question 5: Single Neuron Prediction, Error & Gradient Weight Update

Given:

Input (x) = 4, Weight (w) = 2, Actual Output = 10

Prediction formula: $y_{\text{hat}} = w * x$

Tasks:

- 1) Find the predicted output.
- 2) Calculate the error.
- 3) If the gradient is 4 and learning rate is 0.01, find the updated weight.

1. Problem Inputs

- **Input x :** 4
- **Initial Weight w :** 2
- **Actual Target Output y :** 10
- **Gradient (dE/dw):** 4
- **Learning Rate (η):** 0.01

2. Step-by-Step Computational Process

1) Task 1: Predicted Output Calculation

$$y_{\text{hat}} = w * x = 2 * 4 = 8$$

2) Task 2: Error Calculation

$$\text{Error} = \text{Actual Output} - \text{Predicted Output} = y - y_{\text{hat}} = 10 - 8 = 2$$

$$(\text{Or Squared Loss Error } E = 0.5 * (y - y_{\text{hat}})^2 = 0.5 * (10 - 8)^2 = 2.0)$$

3) Task 3: Weight Update via Gradient Descent

$$\text{Gradient Descent update rule: } w_{\text{new}} = w_{\text{old}} - \eta * \text{gradient}$$

$$w_{\text{new}} = 2 - (0.01 * 4) = 2 - 0.04 = 1.96$$

3. Forward Pass & Gradient Descent Update Flow



4. Summary of Task Results

Final Solution: Predicted Output $y_{\text{hat}} = 8$ | Error = 2 | Updated Weight $w_{\text{new}} = 1.96$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation